1. Introduction

In recent years, quadrotors have garnered significant attention, proving their potential in various fields such as search and rescue, surveillance, and logistics. To successfully perform these missions, the ability to quickly and accurately track reference trajectories that require dynamic and aggressive maneuvers in complex 3D environments is essential.

However, precisely controlling the autonomous flight of a quadrotor involves considerable technical challenges. The quadrotor system has complex characteristics, including strong nonlinear dynamics, coupling between rotors, and unpredictable aerodynamic effects and external disturbances. While the system can be controlled using a linearized model under the small-angle assumption for slow movements like hovering, this assumption is no longer valid during high-speed, aggressive maneuvers, where nonlinear effects become a major, non-negligible factor.

To overcome this nonlinear control problem, various approaches have been attempted over the past decades. Model-based controllers such as Backstepping, Feedback Linearization, and Nonlinear Model Predictive Control (NMPC) have shown high performance by explicitly utilizing the system's dynamic model [Citations needed]. However, these methods have limitations, as they require accurate model parameters, or in the case of NMPC, a high computational cost to solve optimization problems in real-time. On the other hand, data-driven methods like Neural Networks and Reinforcement Learning have been proposed as alternatives to reduce model dependency, but they generally require a vast amount of training data and long training times [Citations needed].

In this paper, to overcome these limitations, we propose a novel hierarchical control architecture that organically combines three control techniques with distinct advantages: **Incremental Nonlinear Dynamic Inversion (INDI), Sliding Mode Control (SMC), and Differential Flatness (DF)**. The proposed controller aims to achieve both high accuracy and robustness in aggressive maneuvering situations through the following hierarchical structure:

- **Outer Loop:** Applies INDI for position and acceleration control. INDI's sensor-based approach ensures robustness against model uncertainties and external disturbances, allowing for the stable generation of a target acceleration.
- **Inner Loop:** Applies SMC for attitude control. The discontinuous control law of SMC guarantees robust tracking of the target attitude despite parameter variations and disturbances during agile maneuvers.
- **Feedforward Control:** Utilizes the differential flatness property of the quadrotor to calculate the desired angular rates and accelerations from the higher-order derivatives (jerk, snap) of the reference trajectory. This feedforward term preemptively reduces the error that the controller must track, maximizing the overall system's tracking performance.

2. System Modeling & Preliminaries

2.1. Quadrotor Dynamics

2.1.1. Translational Dynamics

$$\dot{\mathbf{x}} = \mathbf{v}$$

$$\dot{\mathbf{v}} = g\mathbf{i}_z + \boldsymbol{F}_{thr} \quad + m^{-1}\boldsymbol{F}_{ext}$$

2.1.2. Rotational Dynamics

$$\dot{p} = \frac{1}{J_x}\left[\frac{\sqrt{2}}{2}\cdot(T_2+T_3-T_1-T_4)\cdot l - q\cdot r\left(J_z-J_y\right) - K_2\cdot p\cdot|p|\right] + d$$

$$\dot{q} = \frac{1}{J_y}\left[\frac{\sqrt{2}}{2}\cdot(T_1 + T_3 - T_2 - T_4)\cdot l - p\cdot r(J_x - J_z) - K_2\cdot q\cdot|q|\right] + d$$

$$\dot{r} = \frac{1}{J_z}\left[(Q_1 + Q_2 - Q_3 - Q_4)\cdot l - p\cdot q\left(J_y - J_x\right) - K_2\cdot r\cdot|r|\right] + d$$

$$\dot{\phi} = p + q\cdot\tan\theta\sin\phi + r\cdot\tan\theta\cos\phi$$

$$\dot{\theta} = q\cdot\cos\phi - r\cdot\sin\phi$$

$$\dot{\psi} = q\frac{\sin\phi}{\cos\theta} + r\frac{\cos\phi}{\cos\theta}$$

2.2. Drone Model and Parameters

$$T_{1-4} = T_{max}\cdot v_{1-4}$$

$$Q_{1-4} = Q_{max}\cdot v_{1-4}$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \left(1000\cdot\begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & -1 \end{bmatrix}\cdot\begin{bmatrix} \tau_{thrust} \\ \tau_P \\ \tau_R \\ \tau_Y \end{bmatrix} - P_{min}\right)\cdot\frac{1}{P_{max} - P_{min}}$$

$$\begin{bmatrix} v_1 + v_2 + v_3 + v_4 \\ v_2 + v_3 - v_1 - v_4 \\ v_1 + v_3 - v_2 - v_4 \\ v_1 + v_2 - v_3 - v_4 \end{bmatrix} = \begin{bmatrix} 4\tau_{thrus} & -4 \\ & -4\tau_R \\ & -4\tau_P \\ & 4\tau_Y \end{bmatrix}$$

Parameter	m	l	J_x	J_y	J_z	K_1	K_2
Value	0.8	0.165	0.005	0.005	0.009	0.01	0
Unit	kg	m	kg·m^2	kg·m^2	kg·m^2	N/(m/s)	N·m/(rad/s)
	T_{max}	Q_{max}	P_{max}	P_{min}			
	4	0.05	2000	1000			
	N	N·m	µs	µs			

3. Related Work

3.1. Robust Control for Position and Linear Acceleration Tracking: INDI

The primary reason for applying INDI to the linear acceleration tracker in this study is its high robustness against model errors and disturbances. INDI performs incremental control input calculations based on real-time errors using sensor measurements, which is more practical than traditional model-based nonlinear dynamic control methods. This approach allows for effective control with only minor modifications to the system structure. A key advantage of INDI is that it requires very limited model information for the control system design, and the accuracy of the model used does not significantly impact control performance.

These characteristics have been supported by several studies. The master's thesis by **Eduardo Lima Simoes da Silva** emphasized that INDI requires only small model components and that the control performance is not critically affected even if the accuracy of these components is not perfect [1]. Furthermore, INDI exhibits a characteristic where robustness improves with a higher sampling rate, which provides flexibility in tuning control performance by adjusting the sampling rate. The theoretical foundation is also solid; the work by **Wang et al.** reformulated the INDI control law without the time-scale separation assumption and theoretically analyzed its stability and robustness using Lyapunov theory and nonlinear system perturbation theory. This study proved that the state of the closed-loop system is **ultimately bounded** with respect to a class κ function of the disturbance bounds [2].

This robustness of INDI has been effectively applied, especially in aggressive quadrotor flight control. **Jiesong Yang et al.** utilized INDI's capability to handle nonlinear dynamics and model uncertainty, adding a thrust alignment step to compensate for the mismatch between the actuator and rotational dynamics. This enabled the generation of a smooth polynomial trajectory that satisfies attitude, velocity, and position constraints, achieving both stability and accurate tracking during large-angle maneuvers [3]. Meanwhile, **Zeliang Wu et al.** noted that INDI is gaining attention in aggressive flight research because it can handle nonlinearity and maintain an analyzable structure despite being a sensor-based control method. In their paper, they improved the controller's dynamic responsiveness and reduced its dependency on modeling uncertainty by modeling the thrust unit with high fidelity [4].

Furthermore, there have been active efforts to combine INDI with other techniques to maximize its performance. **Lamsu Kim et al.** combined INDI with a Nonlinear Disturbance Observer (NDO) to more actively respond to the uncertainty term arising from the Taylor expansion while reducing model dependency. This method eliminated steady-state errors and achieved a faster response rate and enhanced robustness [5]. Of particular relevance to this study, the work by **Ezra Tal et al.** effectively handled disturbances and model errors through the incremental control input of the INDI structure, while also generating a feedforward control term that tracks jerk and snap by utilizing the quadrotor's differential flatness property. This controller demonstrated very robust and agile trajectory tracking performance without modeling aerodynamic drag [6].

3.2. Robust Attitude Control for Agile Maneuvers: SMC

Sliding Mode Control (SMC) was adopted to leverage its robustness against various disturbances that can occur during agile maneuvers. In particular, SMC has a strong advantage in guaranteeing stable control even in the presence of model uncertainties or disturbances.

Various approaches have been studied to improve the performance of SMC. **Yutao Jing et al.** enhanced disturbance rejection capability by combining a disturbance observer with each of the SMC position, altitude, and attitude controllers, and optimized the SMC parameters using Particle Swarm Optimization (PSO). They demonstrated superior performance compared to a conventional PID controller [7]. **Dailiang Ma et al.** transformed the under-actuated structure of the quadrotor into a fully-actuated system using differential flatness and then applied a backstepping-based sliding mode controller. They also made the system more robust to disturbances through a state observer. However, this method was based on the assumption that the yaw angle is sufficiently small [8].

Meanwhile, for the practical application of SMC, it is important to solve the persistent chattering phenomenon and the singularity problem of the Euler angle representation. To address these issues, **Hernan Abaunza et al.** adopted quaternion-based rotational dynamics to avoid the singularity problem of the conventional Euler angle approach and effectively reduced the chattering problem by introducing a spherical sliding manifold [9]. **Shikang Lian et al.** also utilized quaternion-based dynamics and eliminated chattering with a Fast Non-Singular Terminal Sliding Mode (FNTSM) that applies a continuous arctangent function. They also compensated for response delay through angular velocity planning [10]. **Ahmed Eltayeb et al.**, from a different perspective, linearized the dynamics through feedback linearization and applied a PID controller, then added SMC to be robust against mass uncertainty. They reduced chattering by applying an error function instead of a switching function in the SMC control law [11].

Recently, attempts to reduce the dependency on the system model have also been gaining attention. **Haoping Wang et al.** combined sliding mode control with a model-free approach to guarantee a bounded error without system parameter information and achieved error elimination within a finite time through SMC. This method demonstrated its performance superiority through comparisons with various existing controllers [12].

4. Differential Flatness

It has been proven that the quadrotor is a **differentially flat system**, where all states and control inputs can be expressed by a set of so-called 'flat outputs' and their time derivatives (Mellinger & Kumar, 2011; Faessler et al., 2017) [13]. The flat outputs for a quadrotor are generally defined as the position of its center of mass and its yaw angle, i.e., $\mathbf{r}_T = [x_\mathbf{T}, y_\mathbf{T}, z_\mathbf{T}, \psi_\mathbf{T}]^T$.

This property is highly significant for controller design. Given a differentiable reference trajectory ($\mathbf{r}_T$), all necessary state variables (e.g., attitude, angular rates) and control inputs (e.g., total thrust) required to follow that trajectory can be calculated algebraically without inverting the complex dynamic equations. In this study, we leverage this property to **generate accurate feedforward control terms**, thereby maximizing the tracking performance for aggressive trajectories.

When a differentiable reference trajectory of the quadrotor's center of mass is given, the desired states and control inputs can be calculated as a function of these flat outputs.

- Reference Trajectory

$$\mathbf{r}_T = [x_\mathbf{T}, y_\mathbf{T}, z_\mathbf{T}, \psi_\mathbf{T}]^T$$
$$\mathbf{v}_T = \dot{\mathbf{r}}_T = [\dot{x}_\mathbf{T}, \dot{y}_\mathbf{T}, \dot{z}_\mathbf{T}, \dot{\psi}_\mathbf{T}]^T$$
$$\mathbf{a}_T = \ddot{\mathbf{r}}_T = [\ddot{x}_\mathbf{T}, \ddot{y}_\mathbf{T}, \ddot{z}_\mathbf{T}, \ddot{\psi}_\mathbf{T}]^T$$

- Reference Attitude

$$
{}^{W}\boldsymbol{R}_{\boldsymbol{B},\boldsymbol{ref}} = \begin{bmatrix} \frac{c\psi_{\mathbf{T}}(\ddot{z}_{\mathbf{T}}+g)^2-(s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}})\ddot{y}_{\mathbf{T}}}{\nu} & -\frac{s\psi_{\mathbf{T}}(\ddot{z}_{\mathbf{T}}+g)}{\eta} & \frac{\ddot{x}_{\mathbf{T}}}{\kappa} \\ \frac{s\psi_{\mathbf{T}}(\ddot{z}_{\mathbf{T}}+g)^2-(s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}})\ddot{x}_{\mathbf{T}}}{\nu} & \frac{c\psi_{\mathbf{T}}(\ddot{z}_{\mathbf{T}}+g)}{\eta} & \frac{\ddot{y}_{\mathbf{T}}}{\kappa} \\ -\frac{(\ddot{z}_{\mathbf{T}}+g)(s\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}}+c\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}})}{\nu} & \frac{s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}}}{\eta} & \frac{\ddot{z}_{\mathbf{T}}+g}{\kappa} \end{bmatrix}
$$

$$
{}^{W}\boldsymbol{R}_{\boldsymbol{B},\boldsymbol{ref}} = [\mathbf{x}_{\boldsymbol{B},\mathbf{T}} \quad \mathbf{y}_{\boldsymbol{B},\mathbf{T}} \quad \mathbf{z}_{\boldsymbol{B},\mathbf{T}}]
$$

Where:

$$
\nu = \sqrt{(\ddot{x}_{\mathbf{T}}^2+\ddot{y}_{\mathbf{T}}^2+(\ddot{z}_{\mathbf{T}}+g)^2)((\ddot{z}_{\mathbf{T}}+g)^2+(s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}})^2)}
$$

$$
\eta = \sqrt{(\ddot{z}_{\mathbf{T}}+g)^2+(s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}})^2}
$$

$$
\kappa = \sqrt{\ddot{x}_{\mathbf{T}}^2+\ddot{y}_{\mathbf{T}}^2+(\ddot{z}_{\mathbf{T}}+g)^2}
$$

$$
\phi_{ref} = \mathrm{atan2}(s\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}-c\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}}, \ddot{z}_{\mathbf{T}}+g)
$$

$$
\theta_{ref} = \mathrm{atan2}(s\psi_{\mathbf{T}}\ddot{y}_{\mathbf{T}}+c\psi_{\mathbf{T}}\ddot{x}_{\mathbf{T}}, \eta)
$$

- Desired Angular Rates

$$
\mathbf{j}_T = [j_{x,\mathbf{T}}, j_{y,\mathbf{T}}, j_{z,\mathbf{T}}, \dddot{\psi}_{\mathbf{T}}]^T
$$

$$
u_1 = m \cdot \sqrt{\ddot{x}_{\mathbf{T}}^2+\ddot{y}_{\mathbf{T}}^2+(\ddot{z}_{\mathbf{T}}+g)^2}
$$

$$
p_{ref} = -\frac{m}{u_1}\left({}^{W}\boldsymbol{R}_{\boldsymbol{B}}(1,2)j_{x,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(2,2)j_{y,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(3,2)j_{z,\mathbf{T}}\right)
$$

$$
q_{ref} = \frac{m}{u_1}\left({}^{W}\boldsymbol{R}_{\boldsymbol{B}}(1,1)j_{x,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(2,1)j_{y,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(3,1)j_{z,\mathbf{T}}\right)
$$

$$
\dot{\phi}_{ref} = \frac{p_{ref} + \dot{\psi}_T \cdot \cos\phi_{ref}\sin\theta_{ref}}{\cos\theta_{ref}}
$$

$$
\dot{\theta}_{ref} = q_{ref} - \dot{\psi}_T \cdot \sin\phi_{ref}
$$

$$
r_{ref} = \dot{\phi}_{ref} \cdot \sin\theta_{ref} + \dot{\psi}_T \cdot \cos\phi_{ref}\cos\theta_{ref}
$$

- Desired Angular Acceleration

$$
\begin{aligned}
\dot{p}_{ref} = -\frac{m}{u_1}\Bigg({}^{W}\boldsymbol{R}_{\boldsymbol{B}}(1,2)s_{x,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(2,2)s_{y,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(3,2)s_{z,\mathbf{T}} + 2p_{ref}\frac{\ddot{x}_T j_{x,T} + \ddot{y}_T j_{y,T} + (\ddot{z}_T+g)j_{z,T}}{\kappa}\Bigg) \\
+ r_{ref} \cdot q_{ref}
\end{aligned}
$$

$$
\begin{aligned}
\dot{q}_{ref} = \frac{m}{u_1}\Bigg({}^{W}\boldsymbol{R}_{\boldsymbol{B}}(1,1)s_{x,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(2,1)s_{y,\mathbf{T}} + {}^{W}\boldsymbol{R}_{\boldsymbol{B}}(3,1)s_{z,\mathbf{T}} - 2q_{ref}\frac{\ddot{x}_T j_{x,T} + \ddot{y}_T j_{y,T} + (\ddot{z}_T+g)j_{z,T}}{\kappa}\Bigg) \\
- r_{ref} \cdot p_{ref}
\end{aligned}
$$

$$A = {}^{W}\boldsymbol{R}_{\boldsymbol{B}}{}^{\boldsymbol{T}} \begin{bmatrix} c\psi_T & -\frac{s\psi_{\mathrm{T}}(\ddot{z}_{\mathrm{T}}+g)}{\eta} & 0 \\ s\psi_T & \frac{c\psi_{\mathrm{T}}(\ddot{z}_{\mathrm{T}}+g)}{\eta} & 0 \\ 0 & \frac{s\psi_{\mathrm{T}}\ddot{x}_{\mathrm{T}} - c\psi_{\mathrm{T}}\ddot{y}_{\mathrm{T}}}{\eta} & 1 \end{bmatrix}, B = {}^{W}\boldsymbol{R}_{\boldsymbol{B}}{}^{\boldsymbol{T}} \begin{bmatrix} -\dot{\psi}_T s\psi_T \\ \dot{\psi}_T c\psi_T \\ \mathbf{0} \end{bmatrix} \dot{\phi}_{ref} + \begin{bmatrix} -r_{ref} \\ 0 \\ p_{ref} \end{bmatrix} \dot{\theta}_{ref}$$

$$\begin{aligned} \ddot{\phi}_{ref} = &\Big(A(2,2)\big(\dot{p}_{ref} - B(1)\big)\Big) \\ &+ \Big(A(1,2)\big(-\dot{q}_{ref} + B(2)\big) + \ddot{\psi}_T\big(A(1,2)A(2,3) - A(1,3)A(2,2)\big)\Big) \\ &/\big(A(2,2)A(1,1) - A(1,2)A(2,1)\big) \end{aligned}$$

$$\begin{aligned} \ddot{\theta}_{ref} = -&\Big(A(2,1)\big(\dot{p}_{ref} - B(1)\big)\Big) \\ &+ \Big(A(1,1)\big(-\dot{q}_{ref} + B(2)\big) + \ddot{\psi}_T\big(A(1,1)A(2,3) - A(1,3)A(2,1)\big)\Big) \\ &/\big(A(2,2)A(1,1) - A(1,2)A(2,1)\big) \end{aligned}$$

$$\dot{r}_{ref} = B(3) + A(3,1)\ddot{\phi}_{ref} + A(3,2)\ddot{\theta}_{ref} + A(3,3)\ddot{\psi}_T$$

5. Controller Overview

5.1. Overall Architecture

The proposed controller has a cascaded structure. In the outer loop, a PD position tracker calculates an acceleration command based on the error, which is then passed to the INDI controller to compute a thrust command vector. The magnitude of this thrust vector represents the required amount of thrust and is sent to the mixer, while the vector's direction indicates the required orientation, or attitude, which is passed to the inner loop.

The inner loop consists of SMC controllers for the roll, pitch, and yaw movements, and a disturbance observer is applied to each to compensate for disturbances. The desired roll ϕ_{des} and pitch θ_{des} are calculated from the desired attitude computed by the INDI controller. The feedforward terms $\dot{\phi}_{ref}$, $\ddot{\phi}_{ref}$, $\dot{\theta}_{ref}$, $\ddot{\theta}_{ref}$, calculated from the reference trajectory, are used to track the desired angular rate and angular acceleration. The yaw controller does not go through the process of the roll and pitch controllers but directly uses ψ_{ref} and its derivatives from the reference trajectory.

5.2. PD Position Tracker

The error terms for position, velocity, and acceleration are defined as:

$$\mathbf{e}_p = \mathbf{r} - \mathbf{r}_T$$

$$\mathbf{e}_v = \dot{\mathbf{r}} - \dot{\mathbf{r}}_T$$

$$\mathbf{e}_a = \ddot{\mathbf{r}} - \ddot{\mathbf{r}}_T$$

The commanded acceleration $\mathbf{a}_c$ is then calculated using a PD control law with an acceleration feedforward term:

$$\mathbf{a}_c = K_p\mathbf{e}_p + K_v\mathbf{e}_v + K_a\mathbf{e}_a + \ddot{\mathbf{r}}_T$$

To control the position and velocity of the vehicle, a PD controller is used. Furthermore, an acceleration feedforward term is incorporated. The reference acceleration, ($\ddot{\mathbf{r}}_T$), from the trajectory is added to the PD output to form the final commanded acceleration, $\mathbf{a}_c$. This ensures that the controller proactively tracks the reference acceleration. The resulting ac is then used as the input for the INDI controller to generate the desired thrust command, F_{des}.

5.3. INDI Controller

5.3.1. Expression for External Force

External force can be represented with acceleration measurement and thrust measurement:

$$\boldsymbol{F}_{ext} = m\left(\mathbf{a}_{meas} - \boldsymbol{F}_{meas}^{thru} - g\mathbf{i}_z\right)$$

$$\mathbf{a}_{\mathbf{tot}} = \boldsymbol{F}_{tot} + g\mathbf{i}_z + m^{-1}\boldsymbol{F}_{ext}$$

$$= \boldsymbol{F}_{tot} + g\mathbf{i}_z + \left(a_{meas} - \boldsymbol{F}_{meas}^{thru} - g\mathbf{i}_z\right)$$

$$= \mathbf{a}_{meas} - \boldsymbol{F}_{meas}^{thrus} + F_{tot}$$

5.3.2. Incremental Thrust Command

The incremental thrust command is given by:

$$\boldsymbol{F}_{des} = \boldsymbol{F}_{meas}^{thrus} + m(\mathbf{a}_c - \mathbf{a}_{meas})$$

$$u_{thrus} = \|\boldsymbol{F}_{des}\|$$

$$\boldsymbol{z}_{B,des} = \frac{\boldsymbol{F}_{des}}{\|\boldsymbol{F}_{des}\|}$$

$$\boldsymbol{z}_{B,des} = \begin{bmatrix} z_{B,des}^1 & z_{B,des}^2 & z_{B,des}^3 \end{bmatrix}^T$$

From the desired body z-axis $\boldsymbol{z}_{B,des} = \begin{bmatrix} z_{B,des}^1 & z_{B,des}^2 & z_{B,des}^3 \end{bmatrix}^T$, the desired roll and pitch angles are derived:

$$\alpha_1 = z_{B,des}^1 \sin\psi_T - z_{B,des}^2 \cos\psi_T$$

$$\alpha_2 = z_{B,des}^1 \cos\psi_T + z_{B,des}^2 \sin\psi_T$$

$$\phi_{des} = \operatorname{atan2}\left(\alpha_1, z_{B,des}^3\right)$$

$$\theta_{des} = \operatorname{atan2}\left(\alpha_2, z_{B,des}^3\right)$$

The INDI controller uses the commanded acceleration, ac, from the higher-level controller to calculate the incremental thrust command, $\boldsymbol{F}_{des}$. The external force term, which includes unknown aerodynamic effects, disturbances, and modeling errors, is implicitly compensated for by using the current measured acceleration in the incremental control law. As a result, the external force term does not appear explicitly in the thrust command equation. This design obviates the need for a separate disturbance observer or other compensation mechanisms for controlling the quadrotor's linear motion.

The calculated thrust command, $\boldsymbol{F}_{des}$, is a 3D vector in the inertial frame that dictates both the magnitude and direction of the required thrust. The magnitude, u_{thrus} , is passed to the motor model to control the speed of each rotor. The direction provides the desired attitude. By normalizing this vector, we obtain the desired body z-axis, $\boldsymbol{z}_{B,des}$. From this unit vector, the desired roll (ϕ_{des}) and pitch (θ_{des}) angles are calculated and passed as references to the inner-loop SMC attitude controller.

5.4. SMC Attitude Controller & Disturbance Observer

The attitude controller not only tracks the desired attitude but also the angular rate and angular acceleration through feedforward terms calculated from the jerk and snap via differential flatness. The goal is to accurately track the jerk and snap by directly utilizing the angular rate and acceleration extracted from them. Additionally, a disturbance observer is added to each controller to estimate disturbances and improve robustness.

5.4.1.Roll Controller

The error terms for the roll controller are defined as follows:

$$e_\phi = \phi - \phi_{des}$$

$$\dot{e}_\phi = \dot{\phi} - \dot{\phi}_{ref}$$

$$\ddot{e}_\phi = \ddot{\phi} - \ddot{\phi}_{ref}$$

Where ϕ_{des} is desired roll angle computed from INDI commanded thrust vector, and $\dot{\phi}_{ref}$ and $\ddot{\phi}_{ref}$ are feedforward term derived from reference trajectory.

According to rotational dynamics:

$$\ddot{\phi} = -\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot\tau_R - \frac{2\sqrt{2}\cdot l\cdot T_{max}\cdot\tan\theta\sin\phi}{J_y}\cdot\tau_p + \frac{4Q_{max}\cdot\tan\theta\cos\phi}{J_z}\cdot\tau_Y + f_\phi$$

Where f_ϕ:

$$f_\phi = q\cdot\left(\frac{1}{\cos^2\theta}\cdot\dot{\theta}\cdot\sin\phi + \dot{\phi}\cdot\tan\theta\cos\phi\right) + r\left(\frac{1}{\cos^2\theta}\cdot\dot{\theta}\cdot\cos\phi - \dot{\phi}\cdot\tan\theta\sin\phi\right) - \frac{J_z - J_y}{J_x}\cdot qr - \frac{K_2}{J_x}\cdot p\cdot|p| - \frac{J_x - J_z}{J_y}\cdot pr\cdot\tan\theta\sin\phi - \frac{K_2}{J_y}\cdot q\cdot|q|\cdot\tan\theta\sin\phi - \frac{J_y - J_x}{J_z}\cdot pq\cdot\tan\theta\cos\phi - \frac{K_2}{J_z}\cdot r\cdot|r|\cdot\tan\theta\cos\phi$$

Define Sliding Surface

$$s_\phi = a_3 e_\phi + \dot{e}_\phi$$

Control Law:

$$\tau_R = k_5\cdot sat\left(\frac{s_\phi}{\epsilon}\right) + k_6\cdot\dot{e}_\phi - \frac{J_x}{J_y}\cdot\tan\theta\sin\phi\cdot\tau_p + \frac{4J_x\cdot Q_{max}\cdot\tan\theta\cos\phi}{2\sqrt{2}J_z\cdot l\cdot T_{max}}\cdot\tau_Y + \frac{J_x}{2\sqrt{2}\cdot l\cdot T_{max}}\cdot\left(f_\phi - \ddot{\phi}_{ref}\right)$$

Stability Analysis:

$$V(s_\phi) = \frac{1}{2}s_\phi^2 \geq 0$$

$$\dot{V}(s_\phi) = s_\phi\cdot\dot{s}_\phi$$

$$\dot{s}_\phi = a_3\dot{e}_\phi + \ddot{e}_\phi = a_3\dot{e}_\phi + \ddot{\phi} - \ddot{\phi}_{ref}$$

$$ref\_\ddot{\phi} = -\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot\tau_R - \frac{2\sqrt{2}\cdot l\cdot T_{max}\cdot\tan\theta\sin\phi}{J_y}\cdot\tau_p + \frac{4Q_{max}\cdot\tan\theta\cos\phi}{J_z}\cdot\tau_Y + f_\phi + \frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot\left(k_5\cdot sat\left(\frac{s_\phi}{\epsilon}\right) + k_6\cdot\dot{e}_\phi\right)$$

$$\ddot{\phi}-\ddot{\phi}_{ref}=-\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot\left(k_5\cdot sat\left(\frac{s_\phi}{\epsilon}\right)+k_6\cdot\dot{e}_\phi\right)$$

$$\dot{s}_\phi=\left(a_3-\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot k_6\right)\dot{e}_\phi-\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot k_5\cdot sat\left(\frac{s_\phi}{\epsilon}\right)$$

If

$$a_3-\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot k_6=0, k_5>0$$

Then,

$$\dot{V}(s_\phi)=-\frac{2\sqrt{2}\cdot l\cdot T_{max}}{J_x}\cdot k_5\cdot\left(s_\phi\cdot sat\left(\frac{s_\phi}{\epsilon}\right)\right)\leq 0$$

Disturbance observer

$$\dot{p}=\frac{1}{J_x}\left[-2\sqrt{2}\cdot T_{max}\cdot l\cdot\tau_R-q\cdot r(J_z-J_y)-K_2\cdot p\cdot|p|\right]+d$$

$$d_{ob}=z_{in}+k_{ob}\cdot p$$

$$\dot{z}_{in}=-k_{ob}\cdot d_{ob}-k_{ob}\cdot\frac{1}{J_x}\left[-2\sqrt{2}\cdot T_{max}\cdot l\cdot\tau_R-q\cdot r(J_z-J_y)-K_2\cdot p\cdot|p|\right]$$

Control Law:

$$\tau_{R,new}=\tau_{R,original}+d_{ob}\cdot\frac{J_x}{2\sqrt{2}\cdot T_{max}\cdot l}$$

5.4.2. Pitch Controller

Similarly, the error terms for the pitch controller are defined:

$$e_\theta=\theta-\theta_{des}$$

$$\dot{e}_\theta=\dot{\theta}-\dot{\theta}_{ref}$$

$$\ddot{e}_\theta=\ddot{\theta}-\ddot{\theta}_{ref}$$

where θ_{des} is desired pitch angle computed from thrust command, and $\dot{\theta}_{ref}$ and $\ddot{\theta}_{ref}$ are computed from jerk and snap of reference trajectory.

According to rotational dynamic equation:

$$\ddot{\theta}=\dot{q}\cdot\cos\phi-q\cdot\dot{\phi}\cdot\sin\phi-\dot{r}\cdot\sin\phi-r\cdot\dot{\phi}\cdot\cos\phi$$

$$=-\frac{2\sqrt{2}\cdot l\cdot T_{max}\cdot\cos\phi}{J_y}\cdot\tau_P-\frac{4Q_{max}\cdot\sin\phi}{J_z}\cdot\tau_Y+f_\theta$$

Where f_θ:

$$f_\theta = -\left(q \cdot \dot{\phi} \cdot \sin\phi + r \cdot \dot{\phi} \cdot \cos\phi + \frac{J_x - J_z}{J_y} \cdot pr \cdot \cos\phi + \frac{K_2}{J_y} \cdot q \cdot |q| \cdot \cos\phi\right) + \frac{J_y - J_x}{J_z} \cdot pq \cdot \sin\phi + \frac{K_2}{J_z} \cdot r \cdot |r| \cdot \sin\phi$$

Next, define sliding surface:

$$s_\theta = a_3 e_\theta + \dot{e}_\theta, a_3 > 0$$

Control law:

$$\tau_P = \frac{1}{\cos\phi} \cdot k_5 \cdot sat\left(\frac{s_\theta}{\epsilon}\right) + \frac{1}{\cos\phi} \cdot k_6 \cdot \dot{e}_\theta - \frac{4 \cdot J_y \cdot Q_{max} \cdot \tan\phi}{2\sqrt{2} \cdot l \cdot T_{max} \cdot J_z} \cdot \tau_Y + \frac{J_y}{2\sqrt{2} \cdot l \cdot T_{max} \cdot \cos\phi} \cdot \left(f_\theta - \ddot{\theta}_{ref}\right)$$

To prove the stability of the control law, let $V(s_\theta) = \frac{1}{2} s_\theta^2 \geq 0$, then:

$$\dot{V}(s_\theta) = s_\theta \cdot \dot{s}_\theta$$

$$\dot{s}_\theta = a_3 \dot{e}_\theta + \ddot{e}_\theta = a_3 \dot{e}_\theta + \ddot{\theta} - \ddot{\theta}_{ref}$$

$$\ddot{\theta}_{ref} = -\frac{2\sqrt{2} \cdot l \cdot T_{max} \cdot \cos\phi}{J_y} \cdot \tau_P - \frac{4Q_{max} \cdot \sin\phi}{J_z} \cdot \tau_Y + f_\theta + \frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot \left(k_5 \cdot sat\left(\frac{s_\theta}{\epsilon}\right) + k_6 \cdot \dot{e}_\theta\right)$$

$$\ddot{\theta} - \ddot{\theta}_{ref} = -\frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot \left(k_5 \cdot sat\left(\frac{s_\theta}{\epsilon}\right) + k_6 \cdot \dot{e}_\theta\right)$$

$$\dot{s}_\theta = \left(a_3 - \frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot k_6\right) \dot{e}_\theta - \frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot k_5 \cdot sat\left(\frac{s_\theta}{\epsilon}\right)$$

And if $a_3 - \frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot k_6 = 0, k_5 > 0$

Then,

$$\dot{V}(s_\theta) = -\frac{2\sqrt{2} \cdot l \cdot T_{max}}{J_y} \cdot k_5 \cdot \left(s_\theta \cdot sat\left(\frac{s_\theta}{\epsilon}\right)\right) \leq 0$$

Disturbance observer

$$\dot{q} = \frac{1}{J_y}\left[-2\sqrt{2} \cdot T_{max} \cdot l \cdot \tau_P - p \cdot r(J_x - J_z) - K_2 \cdot q \cdot |q|\right] + d$$

$$d_{ob} = z_{in} + k_{ob} \cdot q$$

$$\dot{z}_{in} = -k_{ob} \cdot d_{ob} - k_{ob} \cdot \frac{1}{J_y}\left[-2\sqrt{2} \cdot T_{ma} \cdot l \cdot \tau_P - p \cdot r(J_x - J_z) - K_2 \cdot q \cdot |q|\right]$$

Control Law:

$$\tau_{P,new} = \tau_{P,original} + d_{ob} \cdot \frac{J_y}{2\sqrt{2} \cdot T_{ma} \cdot l}$$

5.4.3. Yaw Controller

Unlike the roll and pitch controllers, the desired yaw angle is not derived from the thrust command vector but is given directly by the reference trajectory. Therefore, the yaw controller directly tracks the reference yaw angle, ψ_{ref}, and its derivatives without a separate feedforward term calculation via differential flatness for the attitude itself.

The error terms are defined as:

$$e_\psi = \psi - \psi_{ref}$$

$$\dot{e}_\psi = \dot{\psi} - \dot{\psi}_{ref}$$

$$\ddot{e}_\psi = \ddot{\psi} - \ddot{\psi}_{ref}$$

Rotational dynamics:

$$\begin{aligned}\ddot{\psi} &= \dot{q} \cdot \frac{\sin\phi}{\cos\theta} + q \cdot \frac{\dot{\phi} \cdot \cos\phi \cdot \cos\theta + \dot{\theta} \cdot \sin\phi \cdot \sin\theta}{\cos^2\theta} + \dot{r} \cdot \frac{\cos\phi}{\cos\theta} + r \\ &\quad \cdot \frac{-\dot{\phi} \cdot \sin\phi \cdot \cos\theta + \dot{\theta} \cdot \cos\phi \cdot \sin\theta}{\cos^2\theta} \\ &= \frac{4 \cdot Q_{max} \cdot \cos\phi}{J_z \cdot \cos\theta} \cdot \tau_Y - \frac{2\sqrt{2} \cdot l \cdot T_{max} \cdot \sin\phi}{J_y \cdot \cos\theta} \cdot \tau_P + f_\psi\end{aligned}$$

Where f_ψ:

$$\begin{aligned}f_\psi = \ & q \cdot \frac{\dot{\phi} \cdot \cos\phi \cdot \cos\theta + \dot{\theta} \cdot \sin\phi \cdot \sin\theta}{\cos^2\theta} + r \cdot \frac{-\dot{\phi} \cdot \sin\phi \cdot \cos\theta + \dot{\theta} \cdot \cos\phi \cdot \sin\theta}{\cos^2\theta} - \frac{\cos\phi\,(J_y - J_x)}{J_z \cdot \cos\theta} \\ & \cdot pq - \frac{K_2 \cdot \cos\phi}{J_z \cdot \cos\theta} \cdot r \cdot |r| - \frac{\sin\phi\,(J_x - J_z)}{J_y \cdot \cos\theta} \cdot pr - \frac{K_2 \cdot \sin\phi}{J_y \cdot \cos\theta} \cdot q \cdot |q|\end{aligned}$$

Sliding surface is defined as:

$$s_\psi = a_4 e_\psi + \dot{e}_\psi, a_4 > 0$$

The following control law is:

$$\begin{aligned}\tau_Y = & -\frac{\cos\theta}{\cos\phi} \cdot k_7 \cdot sat\left(\frac{s_\psi}{\epsilon}\right) - \frac{\cos\theta}{\cos\phi} \cdot k_8 \cdot \dot{e}_\psi + \frac{2\sqrt{2} \cdot l \cdot T_{max} \cdot J_z \cdot \tan\phi}{4 \cdot J_y \cdot Q_{max}} \cdot \tau_P - \frac{J_z \cdot \cos\theta}{4 \cdot Q_{max} \cdot \cos\phi} \\ & \cdot \left(f_\psi - \ddot{\psi}_{ref}\right)\end{aligned}$$

To prove the stability, set Lyapunov function to $V(s_\psi) = \frac{1}{2} s_\psi^2 \geq 0$, then:

$$\dot{V}(s_\psi) = s_\psi \cdot \dot{s}_\psi$$

$$\dot{s}_\psi = a_4 \dot{e}_\psi + \ddot{e}_\psi = a_4 \dot{e}_\psi + \ddot{\psi} - \ddot{\psi}_{ref}$$

$$\ddot{\psi}_{ref} = \frac{4 \cdot Q_{max} \cdot \cos\phi}{J_z \cdot \cos\theta} \cdot \tau_Y - \frac{2\sqrt{2} \cdot l \cdot T_{max} \cdot \sin\phi}{J_y \cdot \cos\theta} \cdot \tau_P + f_\psi + \frac{4 \cdot Q_{max}}{J_z} \cdot \left(k_7 \cdot sat\left(\frac{s_\psi}{\epsilon}\right) + k_8 \cdot \dot{e}_\psi\right)$$

$$\ddot{\psi} - \ddot{\psi}_{ref} = -\frac{4 \cdot Q_{max}}{J_z} \cdot \left(k_7 \cdot sat\left(\frac{s_\psi}{\epsilon}\right) + k_8 \cdot \dot{e}_\psi\right)$$

$$\dot{s}_\psi = \left(a_4 - \frac{4 \cdot Q_{max}}{J_z} \cdot k_8\right) \dot{e}_\psi - \frac{4 \cdot Q_{max}}{J_z} \cdot k_7 \cdot sat\left(\frac{s_\psi}{\epsilon}\right)$$

If, $a_4 - \frac{4 \cdot Q_{max}}{J_z} \cdot k_8 = 0, k_7 > 0$:

$$\dot{V}(s_\psi) = -\frac{4 \cdot Q_{max}}{J_z} \cdot k_7 \cdot \left(s_\psi \cdot sat\left(\frac{s_\psi}{\epsilon}\right)\right) \leq 0$$

Disturbance observer

$$\dot{r} = \frac{1}{J_z}\left[4 \cdot Q_{max} \cdot l \cdot \tau_Y - p \cdot q(J_y - J_x) - K_2 \cdot r \cdot |r|\right] + d$$

$$d_{ob} = z_{in} + k_{ob} \cdot r$$

$$\dot{z}_{in} = -k_{ob} \cdot d_{ob} - k_{ob} \cdot \frac{1}{J_x}\left[4 \cdot Q_{ma} \cdot l \cdot \tau_Y - p \cdot q(J_y - J_x) - K_2 \cdot r \cdot |r|\right]$$

Control Law:

$$\tau_{Y,new} = \tau_{Y,original} + d_{ob} \cdot \frac{J_z}{4 \cdot Q_{ma} \cdot l}$$

The control torques, τ_R, τ_P, τ_Y, computed by the respective attitude controllers, provide a unique, closed-form solution. These torques, combined with the total thrust command from the INDI controller, are fed into the mixer to determine the required rotation speed for each rotor.